Answer Key & Teacher Notes

Session 1 — Estimate with Benchmarks · Session 2 — Estimate a Percent of a Number

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages: 8 problems, Find the Error, 2 test items. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Teaching Support Guide	Teacher page: lesson route, what to look for on each exit item, the next prompt, and a fresh check for after reteaching.
Explain Your Thinking	Optional language support for the Guided Notes "Model, Reason, Explain" page: draw it, talk it, write it.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank (English and Spanish), step frames on every problem and larger type (12 pt).

SESSION 1 · TEACHER NOTES

Estimate with Benchmarks

TALK PROMPTS

- Why does naming the whole first change the percent that you estimate?
- How does a benchmark let you judge an answer without doing exact arithmetic?

PARTNER TASK

Elena's Drive. A drive holds 200 GB; 150 GB are used, and 30 GB of that is music. One partner estimates the percent of the drive used (75%); the other estimates the percent of the used space that is music (30 of 150, 20%). Then explain why the wholes differ.

WATCH FOR

- Using the wrong whole (calling 30 GB of music 75% because 150 of 200 GB is used).
- Writing the part as the percent (47 out of 64 called "47%").
- Choosing a benchmark without comparing the part to the benchmark amount (19 out of 60 called 50%).

SESSION 2 · TEACHER NOTES

Estimate a Percent of a Number

TALK PROMPTS

- Why can rounding both the percent and the total still give a useful estimate?
- Does your estimate run high or low? Point to the rounding that tells you.

PARTNER TASK

Tip Estimate and Can Drive Goal. One partner rounds and names the compatible (friendly) numbers; the other builds the estimate from 10% and says whether it runs high or low. Switch roles with Field Trip Coupon: about 19% of $237 (about $48).

WATCH FOR

- Misplacing the decimal when finding 10% (10% of 300 written as 3).
- Multiplying by 20 instead of 0.20 for 20% (20 × 500 = 10,000).
- Thinking any percent of a number is smaller than the number (140% of 195 estimated below 195).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.AT.4	1	2	1	1	2	2	2	2	3	1	1
S1 · Form B	6.AT.4	1	2	1	1	2	2	2	2	3	1	1
S2 · Form A	6.AT.4	1	2	1	2	2	2	2	3	3	1	1
S2 · Form B	6.AT.4	1	2	1	2	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

estimate: A close answer that is not **exact**.

benchmark percent: An **easy** percent to use, like 10%, 25%, 50%, or 75%.

the whole: The full amount. It counts as **100%**.

- First name the **whole**. That amount is 100%.
- Find benchmark amounts (50% = **half**, 25% = one fourth), then pick the one **closest** to the part.

Example: The whole is **80** MB. 50% of 80 MB is **40** MB. 41 MB is just a little more than 40 MB. Answer: About **50%** of the download is done.

YOU TRY

1. a. 25% b. 75% c. 10%

2. GB: 16, 32, 48. 47 is close to 48, so about 75%.

EXIT TICKET

1. a. 50% b. 75%

2. 25% of 300 = 75, and 74 is close to 75, so about 25%.

3. **C** (50%)

PRACTICE FORM A

1. a. 50% b. 25% c. 50% d. 75% e. 10% f. 100%

2. Pages: 60, 120, 180. 118 is close to 120, so about 50%

3. a. $40 b. 75% of $400 = $300, and $310 is close, so about 75%

4. a. 16 cups b. 17 is close to 16, so about 50%

5. a. One fourth of 36 is 9; 27 = 3 × 9, so 75% b. 25%

6. No. 50% of 60 is 30, far from 19. 10% of 60 = 6, so 30% of 60 = 3 × 6 = 18. 19 is close to 18, so 19 out of 60 is about 30%. Also accept about 25% with reasoning (25% of 60 = 15, and 19 is between 15 and 30).

7. Books: 10, 20, 30. a. Between 75% (30 books) and 100% (40 books) b. 75%, because 34 is 4 away from 30 but 6 away from 40

8. a. 10% of 480 = 48, so about 10% b. about 90%

9. 75% describes the used space (150 of 200 GB), not the music. Music is 30 of 200 GB: 10% of 200 = 20 GB, so 30 GB is about 15% of the drive.

10. **B** (one tenth)

11. **B** (25%)

Self-check word **NEAREST**

PRACTICE FORM B

1. a. 50% b. 25% c. 50% d. 75% e. 10% f. 100%

2. Pages: 80, 160, 240. 162 is close to 160, so about 50%

3. a. $60 b. 75% of $600 = $450, and $452 is close, so about 75%

4. a. 24 cups b. 23 is close to 24, so about 50%

5. a. One fourth of 40 is 10; 30 = 3 × 10, so 75% b. 25%

6. No. 50% of 40 is 20, far from 11. 25% of 40 = 10, and 11 is close to 10, so 11 out of 40 is about 25%. Also accept about 30% with reasoning (10% of 40 = 4, so 30% = 12, and 11 is close to 12).

7. Laps: 15, 30, 45. a. Between 50% (30 laps) and 75% (45 laps) b. 75%, because 41 is 4 away from 45 but 11 away from 30

8. a. 10% of 360 = 36, so about 10% b. about 90%

9. 75% describes the used space (300 of 400 GB), not the photos. Photos are 40 of 400 GB: 10% of 400 = 40 GB, so photos are about 10% of the drive.

10. **B** (one half)

11. **B** (25%)

Self-check word **PERCENT**

GUIDED NOTES

estimate: An answer that is **close** to the exact amount.

compatible (friendly) numbers: Numbers that are **easy** to compute with in your head.

ten percent: One tenth of the whole: divide by **10**.

- Round the percent and the whole to **compatible** (friendly) numbers.
- Find **10%** first, then build: 20% is double, 5% is half of 10%.
- Write "about" because the answer is an **estimate**.

Example: 312 is about **300**. 10% of 300 is **30**. 20% is two 10% parts: 30 + 30 = 60. Answer: About **60** bottles

YOU TRY

1. a. about 40 b. about 30 c. about 70

2. Round to $40: 10% = $4, 5% = $2, so about $6.

EXIT TICKET

1. a. about 10 (20% of 50) b. about 30 (75% of 40)

2. 40% of $60: 10% = $6, so 40% = about $24.

3. **C** (80)

PRACTICE FORM A

1. a. about 20 (10% of 200) b. about 30 (20% of 150) c. about 20 (25% of 80) d. about 300 (50% of 600) e. about 27 (30% of 90) f. about 90 (75% of 120)

2. a. 50% of 180 = about 90 lb b. 25% of 180 = about 45 lb c. 10% + 5% of 180 = 18 + 9 = about 27 lb d. 10% of 180 = about 18 lb. Check: 90 + 45 + 27 + 18 = 180. Accept other close estimates with reasoning.

3. Round to $60. 10% = $6, 5% = $3, so the tip is about $9. Accept other close estimates with reasoning.

4. 140% = 100% + 40%. 100% of 200 = 200 and 40% of 200 = 80, so the goal is about 280 cans. It is more than 195 because 140% is more than 100%. Accept other close estimates with reasoning.

5. 20% of $240: 10% = $24, so 20% = about $48. Accept other close estimates with reasoning.

6. No. 9% is about 10%, and 10% of 420 is 42. So 9% of 418 is about 42, not 200 (200 is about half of 418).

7. 75% of 160 = 3 × 40 = about 120 plants. High, because both numbers were rounded up. Accept other close estimates with reasoning.

8. a. 10% of $80 = about $8 b. 4 × $8 = about $32 c. $80 − $32 = about $48; a little high, because $78.90 was rounded up to $80. Accept other close estimates with reasoning.

9. His rounding was fine, but 10% of 300 is 30, not 3. So 20% of 300 = 60; 19% of 285 is about 60 items.

10. **B** (25)

11. **C** (200)

Self-check word **ROUNDED**

PRACTICE FORM B

1. a. about 30 (10% of 300) b. about 50 (20% of 250) c. about 40 (25% of 160) d. about 200 (50% of 400) e. about 18 (30% of 60) f. about 60 (75% of 80)

2. a. 50% of 240 = about 120 lb b. 25% of 240 = about 60 lb c. 10% + 5% of 240 = 24 + 12 = about 36 lb d. 10% of 240 = about 24 lb. Check: 120 + 60 + 36 + 24 = 240. Accept other close estimates with reasoning.

3. Round to $50. 10% = $5, 5% = $2.50, so the tip is about $7.50. Accept other close estimates with reasoning.

4. 130% = 100% + 30%. 100% of 400 = 400 and 30% of 400 = 120, so the goal is about 520 coats. It is more than 410 because 130% is more than 100%. Accept other close estimates with reasoning.

5. 20% of $60: 10% = $6, so 20% = about $12. Accept other close estimates with reasoning.

6. No. 8% is about 10%, and 10% of 500 is 50. So 8% of 512 is about 50, not 400.

7. 75% of 120 = 3 × 30 = about 90 plants. Low, because both numbers were rounded down. Accept other close estimates with reasoning.

8. a. 10% of $60 = about $6 b. 3 × $6 = about $18 c. $60 − $18 = about $42; a little high, because $59.80 was rounded up to $60. Accept other close estimates with reasoning.

9. Lee stopped at 10%. 25% is one fourth, so 25% of 400 = 400 ÷ 4 = 100. 25% of 396 is about 100.

10. **B** (84)

11. **C** (90)

Self-check word **COMPUTE**

SHEET 1 · ESTIMATE WITH BENCHMARKS

1. Pages: 40, 80, 120. 42 is close to 40, so about 25%.
2. a. 10% b. 75% c. 25% d. 50% e. 100% f. 25% g. 10% h. 75%
3. a. $50 b. 20% of $500 = 2 × $50 = $100, and $96 is close to $100, so about 20%. Also accept about 25% with reasoning ($96 is between 10% = $50 and 25% = $125, and closer to $125).
4. 75% of 48 = 36, and 35 is close to 36, so about 75%.
5. 26 out of 50: half of 50 is 25, so it is just over 50% (exactly 52%). 26 out of 60: half of 60 is 30, so it is below 50% (about 43%). 26 out of 50 is closer to 50%.

SHEET 2 · ESTIMATE A PERCENT OF A NUMBER

1. a. about 6 (10% of 60) b. about 30 (75% of 40) c. about 50 (25% of 200)
2. a. $20 b. about $60 (3 × $20)
3. 50% of 400 = 200, so about 200 students.
4. 25% of 800 = 800 ÷ 4 = 200, so about 200 apples.
5. 20% is not 20; she multiplied by 20 instead of 0.20. 10% of 500 = 50, so 20% of 500 = 100. 19% of 512 is about 100.
6. **B** (20)
7. **C** ($48)

SESSION 1 · ESTIMATE WITH BENCHMARKS

Bar: 30, 60, 90 gallons. a. Between 50% (60 gallons) and 75% (90 gallons); 82 is closer to 90, so about 75%. b. 10% of 120 = 12, so 70% = 7 × 12 = 84 gallons. 82 is close to 84, so about 70% (exact: about 68%). c. I found the benchmark amounts of 120 and compared 82 to them. 70% and 75% are both good estimates when the student names the benchmark amount used.

Challenge 82% would mean 82 out of 100, but the whole here is 120, not 100. 82% of 120 is about 98 gallons, much more than 82. 82 is close to 84 = 70% of 120, so the tank is about 70% full.

SESSION 2 · ESTIMATE A PERCENT OF A NUMBER

a. 600 ÷ 4 = about 150 tickets. b. 560 ÷ 4 = about 140 tickets. c. 150 is closer (exact: about 153), because 600 is closer to 588 than 560 is, and 26% is close to 25%. Both are reasonable estimates; give credit for the compatible numbers and the reasoning.

Challenge 150% is more than the whole (100%), so the answer must be more than 58. Round to 60: 100% of 60 = 60 and 50% of 60 = 30, so 150% of 60 = 60 + 30 = 90. 150% of 58 is about 90. 30 is only 50%.

Score each task 0–3: 1 for a model that fits the numbers, 1 for correct math with units, 1 for an explanation that matches the model.